Lesson 1.7 — Inverse Functions

Big idea: An inverse function f^{-1} undoes f. If f sends 3 → 8, then f^{-1} sends 8 → 3. On a graph, inverses are reflections over the line y = x.

Quick review

Algebra method: swap x and y in the equation, then solve for y. That solved equation is $f^{-1}(x)$.

Graph method: reflect the graph of f across the line y = x.

One-to-one check: only functions that pass the horizontal line test have a true inverse function. If f fails it, restrict the domain so it doesn't.

Verification: $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$. If both hold on the proper domain, you have a real inverse.

Worked example

Worked example. Find the inverse of f(x) = 2x – 5.

Write y = 2x – 5. Swap: x = 2y – 5. Solve: y = (x + 5)/2. So $f^{-1}(x) = (x + 5)/2$.
Check: $f(f^{-1}(x)) = 2 \cdot (x + 5)/2 - 5 = x$. ✓

Warm-ups

1. Find the inverse of f(x) = x + 7.
2. Find the inverse of f(x) = 4x.
3. Does $f(x) = x^2$ have an inverse on all real numbers? Why or why not?

Practice

1. Find the inverse of f(x) = 3x – 12.
2. Find the inverse of f(x) = (x + 1) / 5 + 2.
3. Find the inverse of $f(x) = x^3 - 4$.

Challenge

1. Find the inverse of $f(x) = x^2 + 3$ on the restricted domain $x \geq 0$. State the domain of f^{-1}.
2. Are these two functions inverses? f(x) = (x – 1)/2 and g(x) = 2x + 1. Show your check.

Answer key — Lesson 1.7

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $f^{-1}(x) = x - 7$.

2. $f^{-1}(x) = x/4$.

3. No — it fails the horizontal line test (e.g., $f(2) = f(-2) = 4$). You must restrict to $x \geq 0$ (or $x \leq 0$) to get an inverse.

Practice

1. $f^{-1}(x) = (x + 12)/3$.

2. $f^{-1}(x) = 5(x - 2) - 1 = 5x - 11$.

3. $f^{-1}(x) = $ ■$(x + 4)$.

Challenge

1. $f^{-1}(x) = \sqrt{(x - 3)}$. Domain: $x \geq 3$.

2. Yes. $f(g(x)) = (2x + 1 - 1)/2 = x$. $g(f(x)) = 2 \cdot (x - 1)/2 + 1 = x$. ✓